

4.8 L'Hopital's Rule; Indeterminate Forms Worksheet

1. Find $\lim_{x \rightarrow 0} \left(\frac{3x - \sin(x)}{x} \right)$.

Notice that the limit has the indeterminate form of $\frac{0}{0}$. Therefore,

$$\lim_{x \rightarrow 0} \left(\frac{3x - \sin(x)}{x} \right) \stackrel{H}{=} \lim_{x \rightarrow 0} \left(\frac{3 - \cos(x)}{1} \right) = \frac{3 - \cos(0)}{1} = \frac{2}{1} = 2$$

2. Find $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$.

Notice that the limit has the indeterminate form of $\frac{\infty}{\infty}$. Therefore,

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^2} \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{e^x}{2x} \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{e^x}{2} = \infty$$

3. Find $\lim_{x \rightarrow 0} \left(\frac{\sqrt{1+x} - 1 - \frac{x}{2}}{x^2} \right)$.

Notice that the limit has the indeterminate form of $\frac{0}{0}$. Therefore,

$$\lim_{x \rightarrow 0} \left(\frac{\sqrt{1+x} - 1 - \frac{x}{2}}{x^2} \right) \stackrel{H}{=} \lim_{x \rightarrow 0} \left(\frac{\frac{1}{2}(1+x)^{-1/2} - \frac{1}{2}}{2x} \right) \stackrel{H}{=} \lim_{x \rightarrow 0} \left(\frac{-\frac{1}{4}(1+x)^{-3/2}}{2} \right) = \frac{-1/4}{2} = -\frac{1}{8}.$$

4. Find $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\arctan(x)} \right)$.

Notice that the limit has the form of $\infty - \infty$. However we cannot yet apply L'Hopital's rule because we first need the indeterminate form of $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

We note that: $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\arctan(x)} \right) = \lim_{x \rightarrow 0^+} \left(\frac{\arctan(x) - x}{x \arctan(x)} \right)$. This limit has the indeterminate form of $\frac{0}{0}$, so we can apply L'Hopital's rule

$$\lim_{x \rightarrow 0^+} \left(\frac{\arctan(x) - x}{x \arctan(x)} \right) \stackrel{H}{=} \lim_{x \rightarrow 0^+} \frac{\left(\frac{1}{1+x^2} - 1 \right)}{\left(\arctan(x) + \frac{x}{1+x^2} \right)} \stackrel{H}{=} \lim_{x \rightarrow 0^+} \frac{\left(\frac{-2x}{(1+x^2)^2} \right)}{\left(\frac{1}{1+x^2} + \frac{1+x^2 - x(2x)}{(1+x^2)^2} \right)} = \lim_{x \rightarrow 0^+} \frac{\left(\frac{-2x}{(1+x^2)^2} \right)}{\left(\frac{1}{1+x^2} + \frac{1-x^2}{(1+x^2)^2} \right)} = \frac{0}{1+1} = 0$$

5. Find $\lim_{x \rightarrow 0} \left(\frac{1 - \cos(x)}{x + x^2} \right)$.

Notice that the limit has the indeterminate form of $\frac{0}{0}$. Therefore,

$$\lim_{x \rightarrow 0} \left(\frac{1 - \cos(x)}{x + x^2} \right) \stackrel{H}{=} \lim_{x \rightarrow 0} \left(\frac{\sin(x)}{1 + 2x} \right) = \frac{\sin(0)}{1 + 2(0)} = \frac{0}{1} = 0$$

6. Find $\lim_{x \rightarrow \infty} \left(x \tan \left(\frac{1}{x} \right) \right)$.

Notice that the limit has the form of $\infty \times 0$.

$$\lim_{x \rightarrow \infty} \left(x \tan \left(\frac{1}{x} \right) \right) = \lim_{x \rightarrow \infty} \frac{\tan \left(\frac{1}{x} \right)}{\left(\frac{1}{x} \right)}$$

Now the limit is in the indeterminate form $\frac{0}{0}$.

$$\lim_{x \rightarrow \infty} \frac{\tan \left(\frac{1}{x} \right)}{\left(\frac{1}{x} \right)} \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{\sec^2 \left(\frac{1}{x} \right) \left(-\frac{1}{x^2} \right)}{\left(-\frac{1}{x^2} \right)} = \lim_{x \rightarrow \infty} \sec^2 \left(\frac{1}{x} \right) = \sec^2(0) = 1$$

7. Find $\lim_{x \rightarrow \infty} (e^x + x)^{\frac{1}{x}}$.

Notice that the limit has the form of ∞^0 .

$$\lim_{x \rightarrow \infty} (e^x + x)^{\frac{1}{x}} = \lim_{x \rightarrow \infty} e^{\ln \left[(e^x + x)^{\frac{1}{x}} \right]} = e^{\lim_{x \rightarrow \infty} \ln \left[(e^x + x)^{\frac{1}{x}} \right]} = \exp \left(\lim_{x \rightarrow \infty} \frac{1}{x} \cdot \ln (e^x + x) \right)$$

Now the limit has the form $0 \times \infty$.

$$= \exp \left(\lim_{x \rightarrow \infty} \frac{\ln (e^x + x)}{x} \right)$$

Now the limit is in the indeterminate form $\frac{\infty}{\infty}$.

$$\stackrel{H}{=} \exp \left(\lim_{x \rightarrow \infty} \frac{\left(\frac{1}{e^x + x} \right) (e^x + 1)}{1} \right) = \exp \left(\lim_{x \rightarrow \infty} \frac{e^x + 1}{e^x + x} \right) \stackrel{H}{=} \exp \left(\lim_{x \rightarrow \infty} \frac{e^x}{e^x + 1} \right) \stackrel{H}{=} \exp \left(\lim_{x \rightarrow \infty} \frac{e^x}{e^x} \right) = \exp \left(\lim_{x \rightarrow \infty} 1 \right) = \exp (1) = e^1 = e$$

Therefore, $\lim_{x \rightarrow \infty} (e^x + x)^{\frac{1}{x}} = e$.

8. Find $\lim_{x \rightarrow 0^+} x^{\sin(x)}$.

Notice that the limit has the form of 0^0 .

$$\begin{aligned} \lim_{x \rightarrow 0^+} x^{\sin(x)} &= \lim_{x \rightarrow 0^+} e^{\ln(x^{\sin(x)})} = e^{\lim_{x \rightarrow 0^+} \ln(x^{\sin(x)})} = \exp \left(\lim_{x \rightarrow 0^+} \ln(x^{\sin(x)}) \right) = \exp \left(\lim_{x \rightarrow 0^+} \sin(x) \cdot \ln(x) \right) = \exp \left(\lim_{x \rightarrow 0^+} \frac{\ln(x)}{\left(\frac{1}{\sin(x)} \right)} \right) \\ &= \exp \left(\lim_{x \rightarrow 0^+} \frac{\ln(x)}{\csc(x)} \right) \end{aligned}$$

Now we have the limit in the indeterminate form $\frac{-\infty}{\infty}$.

$$\begin{aligned} \stackrel{H}{=} \exp \left(\lim_{x \rightarrow 0^+} \frac{\left(\frac{1}{x} \right)}{-\csc(x) \cot(x)} \right) &= \exp \left(\lim_{x \rightarrow 0^+} \frac{-\sin(x) \tan(x)}{x} \right) \stackrel{H}{=} \exp \left(\lim_{x \rightarrow 0^+} \frac{-\cos(x) \tan(x) - \sin(x) \sec^2(x)}{1} \right) \\ &= \exp \left(-\cos(0) \tan(0) - \sin(0) \sec^2(0) \right) = \exp \left((-1)(0) - (0)(1) \right) = \exp(0) = e^0 = 1 \end{aligned}$$

Therefore, $\lim_{x \rightarrow 0^+} x^{\sin(x)} = 1$.